

Computer Engineering 1

Course Organization and Examination Conditions

Course Structure

- **2 Theory Lessons**
 - Universal Knowledge
 - Preparation for Lab Sessions
- **2 Lab Lessons**
 - Assignments Assessed
 - On-site Submission
- **2 Lessons Self-Study**
 - Preparation / Review
 - Completion of Lab Work
 - Approx. 2 Hours / Week

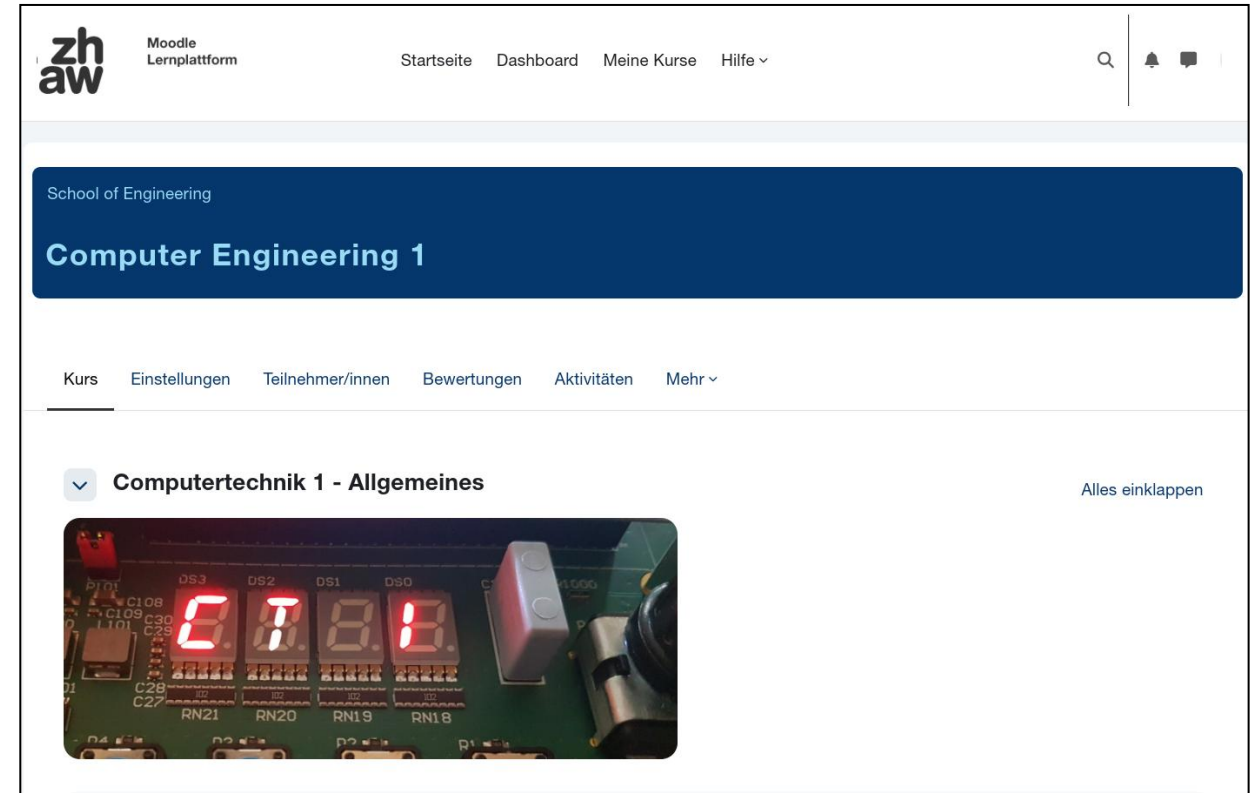
Course Content → Details in Moodle

	Theory – 2 Lectures	Labs – 2 Lektures
1	Introduction Computer Engineering	CT Board and Development Environment (C)
2	Cortex-M Architecture	Bit Manipulations (C)
3	Data Transfer	Introduction to Assembly
4	Arithmetic Operations	Data Transfer Commands
5	Arithmetic Operations (continuation) / Casting	Arithmetic Operations
6	Logic and Shift/Rotate Instructions / Branches	ALU and Branch Instructions
7	Control Structures	Control Structures
8	Subroutines and Stack	Structured Programming in Assembly
9	Written Exam (45 Minutes) / Reserve	Structured Programming in Assembly (continuation)
10	Parameter Passing	Subroutines and Parameter Passing
11	Modular Coding and Linking	Linker
12	Exceptional Control Flow – Interrupts	Multiplication
13	Exceptional Control Flow – Interrupts (continuation)	Interrupt
14	Increase Processor Efficiency	Finish labs / questions and answers

Course Materials and Information Moodle

All information available via Moodle:

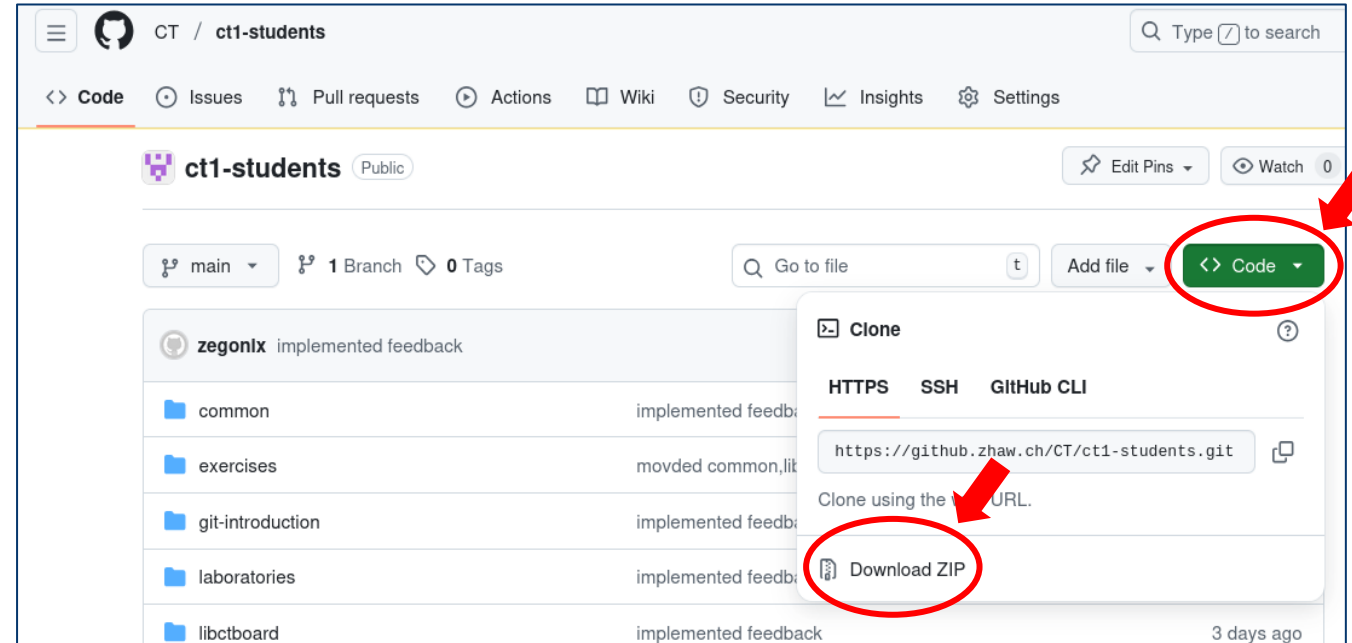
- Teaching materials (links to GitHub)
 - Lecture slides
 - Exercises for lectures
 - Lab assignments and materials
 - Student quiz (directly in Moodle)
- Course information
 - Semester schedule
 - Examination conditions
 - Learning materials



<https://moodle.zhaw.ch/>

... Course Materials and Information Github

- Files are hosted on GitHub (Moodle only contains links)
- Download to local computer
 - Download as ZIP or
 - Clone using git
 - **Do not create a fork!**
(160 forks become difficult to manage...)



<https://github.zhaw.ch/CT/ct1-students>

CT-Board

Provided on loan for CT1 and CT2

- STM32F4 Microcontroller
- Peripherals

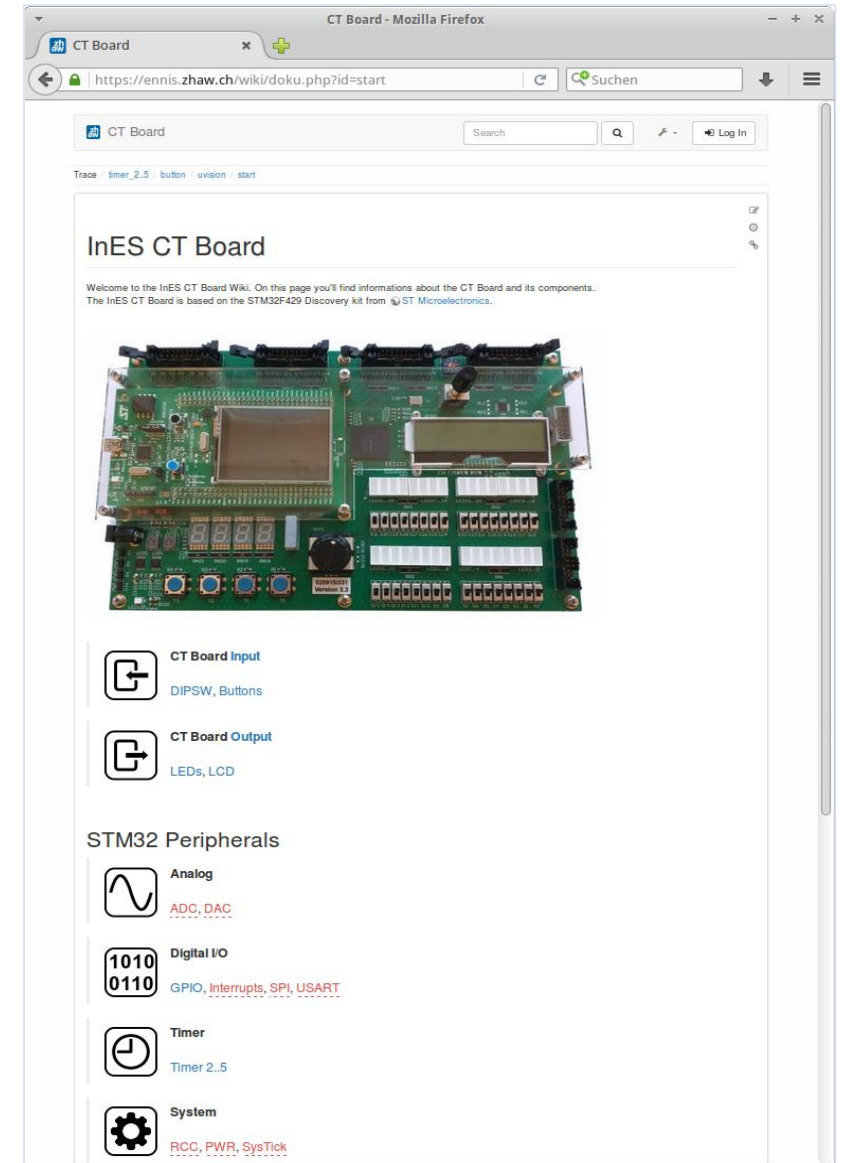


CT-Board

CT Board Wiki

- **Technical Documentation**
 - CT Board
 - STM32 Microcontroller
- **Software**
 - InES uVision Pack for Keil
 - Keil Installation Guide

remark: if you want to use a different IDE than Keil, please refer to the installation guide in Github



<https://ennis.zhaw.ch/>

Assessment

Graded Components

– Lab assessment	15%
– Midterm examn	15%
– Final examn	70%

Assessment Labs (15%)

- 14 Lab Sessions
- Completion of lab work through self-study (follow-up work)
- A working solution must be demonstrated to the lecturer no later than the beginning of the next lab session.
- In case of absence (illness), the solution must be submitted later. Exceptions must be agreed in writing with the lecturer.
- You must be able to explain your solutions and answer related questions.
- Assessment criteria and point distribution can be found in the respective lab description.

$$Note = \frac{\textit{erzielte Punktzahl} \cdot 5}{\textit{max. mögliche Punktzahl}} + 1$$

Assessment Examinations

- **Midterm Exam (15%)**
 - Duration: 1 Lesson (45 Minutes)
- **Final Exam (70%)**
 - Duration: 2 Lessons (90 Minutes)
- **Examination Conditions**
 - The content covered in theory and labs will be examined
 - Written exam, according to lecturer
 - Permitted Aids
 - 20 A4 pages (or 10 double-sided) self-written summary
 - Calculator without communication capabilities

Let's Get Started!

